Coarticulatory propensity in Khalkha Mongolian

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Coarticulatory propensity and directionality in non-harmonic VCV sequences

Purpose: Measuring coarticulatory directionality and propensity in non-harmonic sequences in Khalkha Mongolian

Introduction

- Continuous acoustic variation in speech due to overlapping articulatory gestures [1].
- Vowel-to-Vowel (V-to-V) coarticulation is a special case of non-contiguous coarticulation resulting from anticipatory articulatory planning on the one hand, and carryover coarticulation on the other.
- Uniform way of understanding the variation has been to model the dynamic and kinematic differences in articulatory overlap. Differences have been modelled variously in space and time [2, 3].
- In V_1CV_2 sequences:
  1. The acoustics of V_1 are perturbed by the anticipatory production planning of V_2 gestures
  2. The acoustics of V_2 are perturbed by the mechanico-inertial properties of articulatory gestures of V_1
- Coarticulatory acoustic variation in V-to-V sequences once phonologized are known to provide the conditioning for the development of vowel harmony (VH) patterns [4, 5].
- Typically, in non-harmonic languages, acoustic variation resulting from coarticulation is perceptually compensated by listeners.
- Lack of such perceptual compensation has been shown to be responsible for the development of VH patterns [6].
- Unlike, suffixal harmony systems like Assamese [7], Bengali [8], and Telugu [9, 10] where the direction of the harmony is right-to-left, Khalkha, primarily, is a stem internal harmony system where the direction of the harmony proceeds from left-to-right.

♪ Our goals

1. Examine the magnitude and extent of V-to-V coarticulation in Khalkha Mongolian, which exhibits both Advanced Tongue Root [ATR] harmony and rounding harmony.
2. Examine the effect of formants from V_1 on V_2 and vice-versa using a set of linear fixed effect models.

- **What we find:** The direction of coarticulation in non-harmonic sequences is greater in the anticipatory direction, opposite to the direction of suffixal VH in Khalkha Mongolian.
- **What we find:** Effect only significant for F_1 values and not F_2; coarticulatory formant perturbation runs in the tongue height dimension and not tongue front-back dimension.

Coarticulatory propensity

- Coarticulatory propensity is the spatial and temporal extent of coarticulation between or across segments.
- Coarticulatory resistance of segments with variable articulatory gestures affects acoustic variation [11, 12].
- Nature of contrast affects the magnitude of coarticulation in V-to-V sequences, in languages that exhibit VH of the type where distinctive features are shared between vowels [13] and in those where VH results in vowel copying [14].

Corpus and measures

Speech corpus Carrier phrase: "pi <target word/distractor> gesen". Recordings were conducted in a quite environment in The EFL University.

- 4 female speakers; 59 phrases, with 4 iterations from each speaker

Annotation Manual annotation in Praat [15]

- Single-tier annotation
- Vowel identity with contextual information

Measurements 10 vowel formants from V_1CV_2 sequences

- Lobanov normalized formants to eliminate specific speaker effects [16].

VH and Khalkha vowel inventory

- Vowel harmony is a phonological process that restricts the co-occurrence of vowels (usually) within a non-compound word

Vowel Inventory *i, e, a, o, ɔ, u and ʊ*

Non-initial archiphonemes /E/ and /U/ have missing features that get filled in through vowel harmony

VH in Khalkha Mongolian

Harmony types [ATR] and Rounding

- [ATR] Harmony
  - Distinguishes tense and lax vowels in Mongolian, which Svantesson (2005) refers to as 'pharyngeal' and 'non-pharyngeal'
  - Vowels in non-compound words must share values for [+/- ATR], depending on the root/stem vowel (at the morpho-phonemic level).

Trigger vowels: *o, ɔ, u, ʊ, a, e* (in initial positions)

Target vowels: Archiphonemes /E/ and /U/

Alternations:

/E/: /e/, /a/, /o/, or /ɔ/| /U/: /u/, or /ʊ/

- Rounding Harmony
  - Influences vowels to surface as rounded when the neighbouring vowel (the root/stem vowel for Khalkha Mongolian) is rounded
  - *T*he trigger must be nonhigh.
  - *T*he trigger and target must agree in height.

Trigger vowels: /o/ and /ɔ/

Target vowels: Archiphoneme /E/

- Non-initial *i* is transparent
- *u and ʊ* are opaque (restricted to rounding harmony)

F_1 and F_2 plots on X-Y euclidean plane

F_1 and F_2 plots on X-Y euclidean plane (PhonR package [17]).

- Formant data visualizations give the relative positions of the vowels and the areas of the vowel polygons for V_1 and V_2.
- Left panel represents steady-state positions and the vowel area polygons for V_1 and right panel represents the same for V_2.
- **Observations**
  1. The mid-front vowel *e* is raised and is overlapped significantly with the high front vowel *i*
  2. The *e* is lower than the *i*
- F_1 and F_2 values of *e* are perturbed by the high vowel in the V_2 position; implies a stronger anticipatory effect, at least for the vowel *e*.

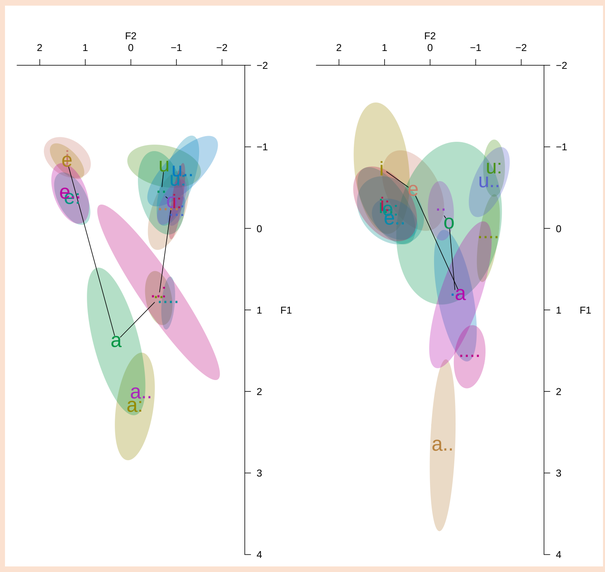


Figure 1: Steady state distal formants: Vowel ellipses and polygons from V_1T6, V_1 mid-point (left) and V_2T_5; V_2 mid-point (right)

- Greater overlapping effect due to the presence of a consonant, yet the polygon area of V_1 is larger than that of V_2.
- **Observation**
  - The effect of V_2 on V_1 is robust across their different positions in time (vowel-mid position, onset and offset).

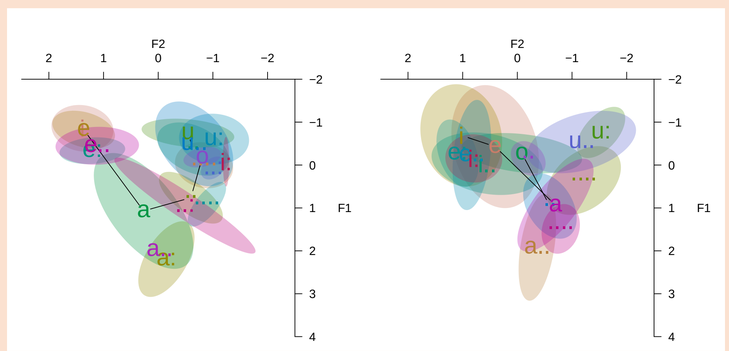


Figure 2: Proximal formants: Vowel ellipses and polygons from V_1T6, V_1 mid-point (left) and V_2T_5; V_2 mid-point (right)

Vowel polygon areas

Vowel polygon area for V_1 is larger than V_2

Polygon Area	Subject f1	Subject f2	Subject f3	Subject f4
Distal				
V1 at T6	**3.48591494**	**4.1172744**	**2.5307905**	**2.8948019**
V2 at T5	0.09251035	0.6124816	0.1368078	0.3408551
Proximal				
V1 at T10	**2.725822**	**3.827627**	**1.998709788**	**2.1165014**
V2 at T1	0.3790905	0.536976	0.008869308	0.1819477

Table 1: Vowel polygon areas of V_1 and V_2 from distal and proximal formant measures

Statistical analyses

- Linear mixed effects models tested the variable and mutual effect of F_1 and F_2 values from distal and proximal V positions to quantify coarticulatory propensity in non-harmonic V_1CV_2 sequences.
- **Model 1**: effect of F_1 from T5 of V_2 and a fixed effect of V_2 on the F_1 from V_1 at T6. This model returns a t=3.543 with no significant effect of V_2.
- **Model 2** effect of F_1 from T6 of V_1 and a fixed effect of V_1 on the F_1 from V_2 at T5. This model returns a t=3.233 and a t=2.185 for vowel V_1, /u:/
- **Main effect**: Significant covariation of F_1 values with slightly greater effect of V_2 on V_1; **directionality of coarticulation (F_1) is in the anticipatory direction**, opposite to the direction of the suffixal harmony system.
- Similar models on F_2 do not show significant effects, with t=-0.044 and t=-0.020 for F_2 of V_1 at T6 and F_2 of V_2 at T5

Conclusions

- Stem-internal and stem + suffix harmonic sequences in Mongolian exhibit a left-right directionality.
- Lack of perceptual compensation [6] is known to contribute to the development of VH systems, to the extent that it might regress to a radical case of vowel-copying, such as in Telugu [14, 10, 18].
- Coarticulatory propensity is greater in the anticipatory direction in non-harmonic Khalkha Mongolian sequences
- Opposite to the direction of both ATR and rounding harmony to maintain contrast, where harmonic sequences may lead to contrast obliteration through featural neutralization.
- The directionality of coarticulatory propensity suggests that articulatory planing seeks to preserve contrast.
- Unlike the intuitive notion of coarticulation as a force that contrives against contrast, Mongolian shows that coarticulation augments contrast.
- V-to-V coarticulation patterns in non-harmonic sequences seek to enhance contrast by providing perceptual advantage through variable coarticulatory propensity.

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